

PUNTA DEL ESTE, Uruguay

WMO: 865860

Lat: 34.860S

Lon: 55.097W

Elev: 35

StdP: 100.91

Time Zone: -3.00 (W03)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.3	3.9	-0.2	3.7	7.0	0.1	3.8	7.2	13.2	9.5	11.6	10.3	1.7	300	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.7	32.0	22.0	30.2	21.4	28.9	21.0	24.0	28.6	23.2	27.3	22.4	26.3	5.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	17.7	25.9	22.0	16.7	25.1	21.1	15.9	24.2	72.5	28.4	69.1	26.7	66.3	26.5	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.3	9.3	DB	-0.1	36.0	1.3	1.3	-1.0	36.9	-1.7	37.6	-2.4	38.3	-3.3	39.2	(4)
(5)			WB	-0.2	26.6	1.1	1.2	-1.1	27.4	-1.7	28.1	-2.4	28.8	-3.2	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	22.8	22.5	20.5	17.6	14.4	11.7	10.9	12.3	13.5	16.0	18.5	21.0	(6)	
	DBStd	5.15	2.74	2.66	2.68	3.09	3.10	3.11	3.48	3.64	3.25	3.00	3.23	3.12	(7)	
	HDD10.0	67	0	0	0	0	2	14	29	15	6	1	0	0	(8)	
	HDD18.3	1102	2	2	11	49	127	203	232	192	151	86	36	12	(9)	
	CDD10.0	2536	398	349	326	228	138	64	57	85	109	186	254	342	(10)	
	CDD18.3	530	141	117	78	27	5	2	2	4	4	13	41	95	(11)	
	CDH23.3	3272	994	698	347	88	17	8	7	33	28	86	286	680	(12)	
(13)	CDH26.7	941	322	203	68	12	1	1	1	9	7	16	79	223	(13)	
(14) Wind	WSAvg	4.5	4.7	4.7	4.4	4.2	3.9	4.1	4.3	4.5	5.0	4.9	4.9	4.9	(14)	
(15) Precipitation	PrecAvg	1154	83	103	113	113	94	108	102	80	84	108	96	81	(15)	
	PrecMax	1568	147	241	399	241	258	216	182	158	159	264	179	209	(16)	
	PrecMin	703	19	26	18	5	5	42	26	14	14	17	28	14	(17)	
	PrecStd	229	36	60	90	60	54	49	47	40	39	64	44	55	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.3	33.8	30.9	28.2	25.7	24.2	24.1	27.9	26.9	28.8	31.8	33.1	(19)	
		MCWB	22.5	22.7	22.2	21.1	18.9	19.7	18.3	20.0	20.0	19.7	20.7	21.8	(20)	
	2%	DB	31.9	31.0	28.7	25.8	22.4	20.8	20.1	22.8	23.0	25.8	29.0	31.1	(21)	
		MCWB	22.3	22.4	21.2	19.2	17.4	17.2	16.1	16.9	16.5	18.4	19.8	20.8	(22)	
	5%	DB	30.0	29.1	26.9	24.0	20.5	18.1	17.7	20.0	20.9	23.2	26.8	29.1	(23)	
		MCWB	21.7	21.9	20.8	19.1	16.7	15.5	14.9	15.9	15.5	17.1	19.1	20.4	(24)	
	10%	DB	28.2	27.2	25.1	22.3	19.0	16.2	15.9	17.9	18.9	21.2	24.2	27.1	(25)	
		MCWB	21.2	21.2	20.2	18.7	16.2	13.6	13.6	14.5	14.8	16.5	18.1	19.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	25.0	24.2	22.6	20.2	20.1	19.1	19.8	20.7	21.7	22.6	24.5	(27)	
		MCDB	29.9	30.2	27.9	25.8	23.3	23.3	22.6	25.9	25.1	26.1	28.6	30.2	(28)	
	2%	WB	24.1	23.8	22.9	21.4	18.7	18.0	16.9	17.9	18.4	19.7	21.1	22.7	(29)	
		MCDB	28.7	28.2	25.9	23.7	21.0	19.9	19.5	21.5	21.3	23.2	25.8	27.3	(30)	
	5%	WB	23.1	23.0	22.0	20.5	17.6	16.2	15.4	16.6	16.7	18.5	20.1	21.7	(31)	
		MCDB	27.2	27.2	24.9	22.5	19.6	17.7	17.2	19.1	19.1	21.5	24.4	26.2	(32)	
	10%	WB	22.3	22.2	21.1	19.5	16.6	14.4	14.1	15.2	15.5	17.5	19.1	20.8	(33)	
		MCDB	26.2	25.8	23.9	21.7	18.5	15.9	15.7	17.2	17.9	20.4	22.9	25.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	8.7	8.5	8.3	7.9	8.0	7.8	8.2	8.4	8.7	9.7	9.8	(35)	
		MCDBR	12.5	11.2	10.6	10.1	9.5	9.8	9.8	11.8	12.2	12.3	13.2	13.2	(36)	
	5% WB	MCWBR	6.1	5.5	5.4	5.6	5.8	6.7	6.5	7.2	6.9	6.7	6.6	6.3	(37)	
		MCWBR	10.2	9.5	8.1	7.8	7.5	8.4	8.6	10.1	9.6	9.8	10.8	10.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.392	0.391	0.370	0.365	0.355	0.362	0.361	0.443	0.454	0.396	0.381	0.384	(40)	
		taud	2.430	2.453	2.500	2.462	2.464	2.421	2.397	2.106	2.098	2.336	2.408	2.424	(41)	
	Ebn at Noon		944	923	906	849	799	757	780	747	805	907	948	956	(42)	
		Edh at Noon	122	115	102	94	83	81	87	132	150	128	124	124	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.28	6.23	5.19	3.78	2.63	2.21	2.31	3.05	4.23	5.52	6.82	7.45	(44)	
	RadStd		0.52	0.57	0.44	0.41	0.18	0.20	0.22	0.30	0.31	0.47	0.54	0.49	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	+0.48	+0.70	N/A	+0.69	+0.63	N/A	N/A	N/A	+125	+67

Nomenclature: See separate page